

4D Time Geometry in General Relativity

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Frequency Gradient Distributions in 3D Physical Space

**Understanding geometric time and gravitational time dilation in general relativity:
an intuitive approach**

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This article provides an introduction to the theory of time I previously presented in my preprint.

Abstract: This article presents an intuitive reinterpretation of the temporal geometry of general relativity (GR) using a physically concrete three-dimensional model. We conceptualize space as a network of sites, each endowed with a harmonic oscillator degree of freedom whose resonant frequency directly determines the rate of proper time. Within this framework, the abstract 4D time geometry of GR is precisely equivalent to the frequency gradient distribution across these spatial sites. Gravitational time dilation and redshift emerge naturally as manifestations of a non-uniform frequency field. By enforcing a topological constraint of 3D spatial invariance, this model inherently avoids singularities and provides a unified, operationally clear picture where spacetime curvature, energy, and frequency become directly interlinked.

The 4D formulation of General Relativity is abstract, while human cognition is inherently 3D. This makes it difficult to intuitively grasp the geometric time curvature in GR.

Here I provide an intuitive picture:

Stronger Constraint than 4D Diffeomorphism Invariance: Let us start from the perspective of 3D real space. GR assumes 4D diffeomorphism invariance. Here, we impose a stronger constraint: 3D topological invariance — meaning the adjacency relations between spatial sites remain fixed. Under this setting, topologically singular solutions (i.e., singularities) cannot exist.

Time Curvature as a Distribution of Resonant Frequencies: Each spatial site possesses a resonant degree of freedom (as a simplified approximation analogous to a phonon system; in reality, the degrees of freedom of a spatial site include not only resonance but also spin and axial rotation). The resonant frequency of each site is precisely the rate at which proper time flows at that site. Consequently, the spatial distribution of resonant frequencies across all sites is the time curvature of GR.

Continuum Limit: GR is formulated as a continuous field. This discrete resonant system approximates that continuous field in the appropriate limit.

Mirror Relation: Based on gravitational redshift, cosmological expansion redshift, and the classical gravitational time dilation relations of GR, we can establish, within this model, a mirror relation between 3D geometric curvature and the resonant frequencies of spatial sites. In this way, the abstract geometric time curvature of GR is concretely represented as a frequency distribution across spatial sites.

The Nature of Electromagnetic Waves and Energy: Within this physical model, an electromagnetic wave is essentially a transverse wave propagating through the spatial site network — sites that possess spin, resonant, and other degrees of freedom. The frequency of the electromagnetic wave is exactly the frequency of the spatial sites. And since the frequency of an electromagnetic wave is directly related to its energy ($E=h\nu$), the frequency of a spatial site also directly represents energy.

In this way, all the geometric-dynamic effects associated with the time component in general relativity can be fully represented by the scalar frequency field of this three-dimensional harmonic oscillator system and its gradient. The temporal geometry in the 4D formalism of general relativity corresponds precisely to the frequency gradient distribution of the spatial sites in 3D space.

Conclusion for the reader: Do you see it now? The direct relationship between the time geometry of relativity and energy becomes transparent.

Does this picture clarify the abstract 4D Riemannian geometry by reducing it to intuitive 3D geometry plus a frequency distribution over space?

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Topology-Preserving Configuration. <https://doi.org/10.5281/zenodo.15278220>